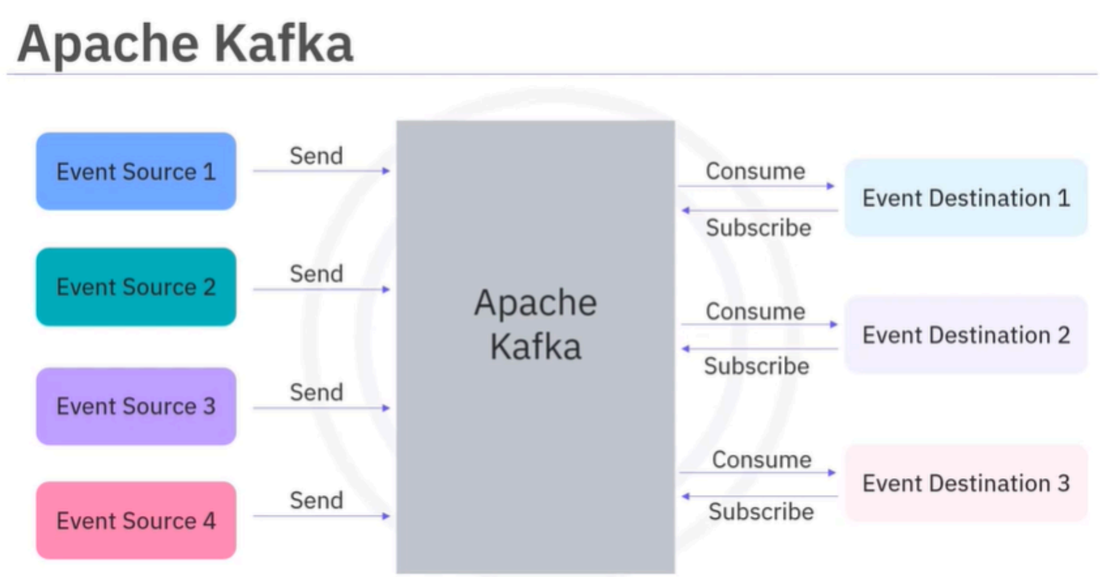
Apache Kafka Overview

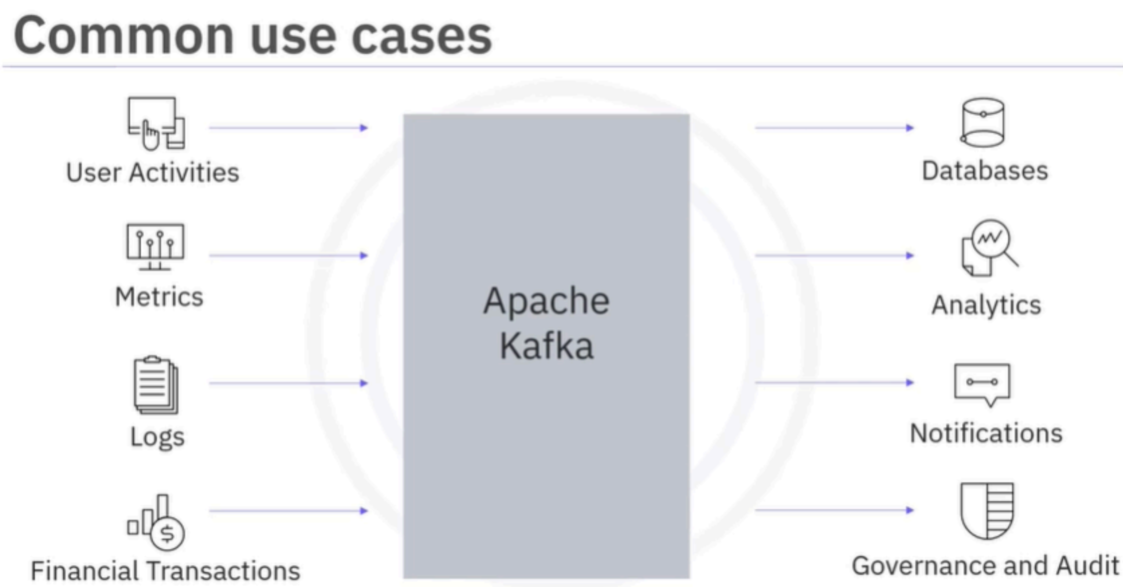
Sure! Let's talk about Apache Kafka, which is an event streaming platform.

Apache Kafka is like a busy post office that helps send and receive messages between different places. Imagine you have a lot of friends who want to share their thoughts and experiences with each other. Instead of each friend trying to talk directly to everyone else, they send their messages to the post office (Kafka). The post office collects all these messages and then delivers them to the right friends when they are ready to receive them. This way, everyone can communicate efficiently without missing out on important information.

Kafka is used in many situations, like tracking user activities on websites, monitoring sensors, or processing financial transactions. It helps businesses manage large amounts of data quickly and reliably, ensuring that important messages are stored safely and can be accessed whenever needed.

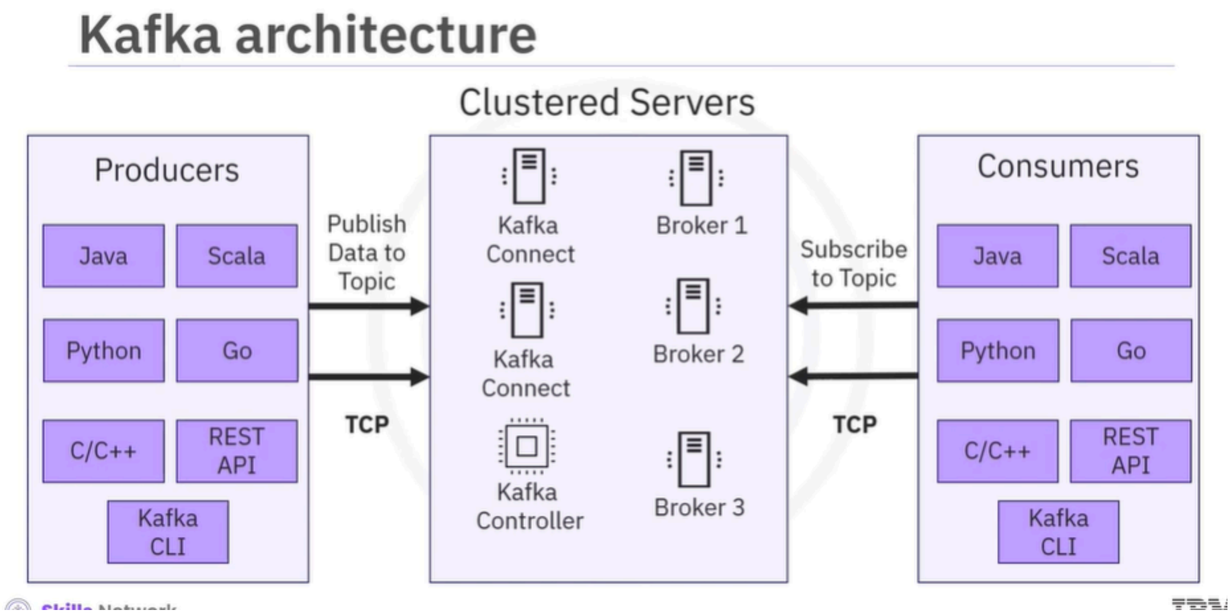


- Implementing an ESP and its components from scratch can be extremely difficult, but there are several open source and commercial ESP solutions with built-in capabilities available in the market. Apache Kafka is an open source project which has become the most popular ESP.



- Kafka is a comprehensive platform and can be used in many application scenarios. Kafka was originally used to track user activities such as keyboard strokes, mouse clicks, page views, searches, gestures, screen time, and so on. But now Kafka is also suitable for all kinds of metric streaming such as sensor readings, GPS, and hardware and software monitoring.
- For enterprise applications and infrastructure with a huge number of logs, Kafka can be used to collect and integrate them into a centralized repository. For banks, insurance, or fintech companies, Kafka is widely used for payments and transactions.
- These scenarios are just the tip of the iceberg. Essentially, you can use Kafka when you want high throughput and reliable data transportation services among various event sources and destinations. All events will be ingested in Kafka and become available for subscriptions and consumption, including further data storage and movement to other online or offline databases and backups.

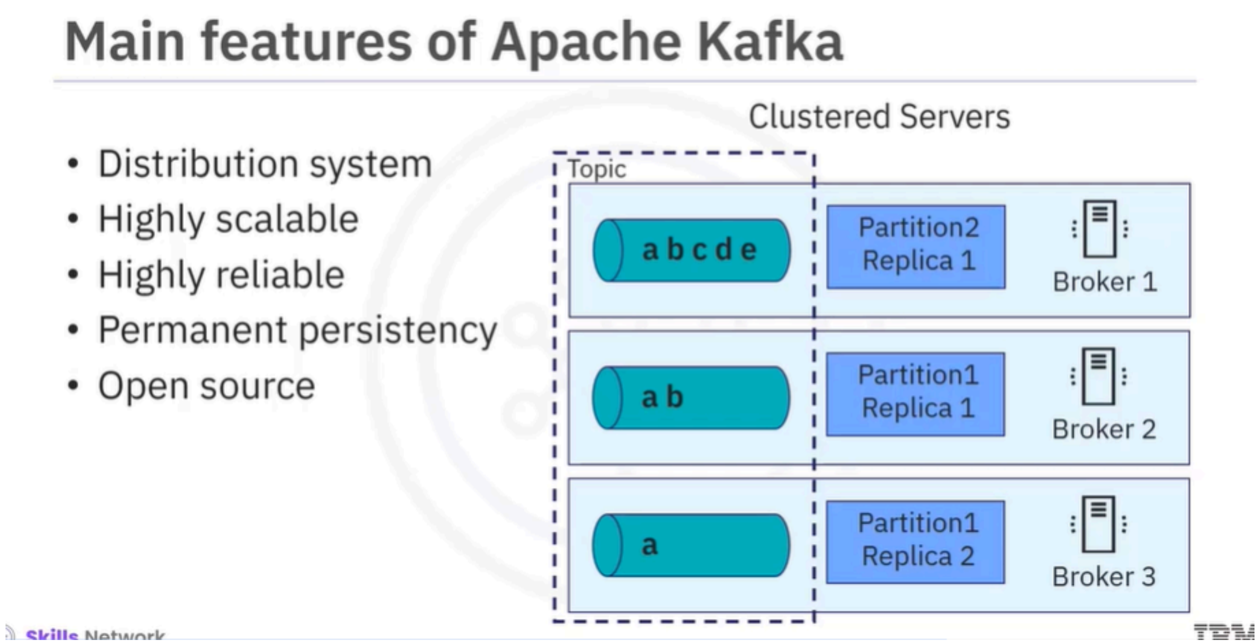
- Real time processing and analytics including dashboard, machine learning, AI algorithms, and so on, generating notifications such as email, text messages, and instant messages, or data governance and auditing to make sure sensitive data such as bank transactions are complying with regulations.



- Kafka is a distributed, real time event streaming platform that adheres to client server architecture. Kafka runs as a cluster of broker servers, acting as the event broker to receive events from the producers, store the streams of records, and distribute events. It also has servers that run Kafka Connect to import and export data as event streams. Please note that all the brokers before version 2.8 relied on another distributed system called Zookeeper for management and to ensure all brokers work in an efficient and collaborative way. However, Kafka Raft, pronounced as KRaft, is now used to eliminate Kafka's reliance on Zookeeper for metadata management. It is a consensus protocol that streamlines Kafka's architecture by consolidating metadata responsibilities within Kafka itself.
- Using Kafka controllers, producers send or publish data to the topic, and the consumers subscribe to the topic to receive data. Kafka uses a transmission control protocol, TCP based network communication protocol, to exchange

data between clients and servers. For the client side, Kafka provides different types of clients such as Kafka command line interface, CLI.

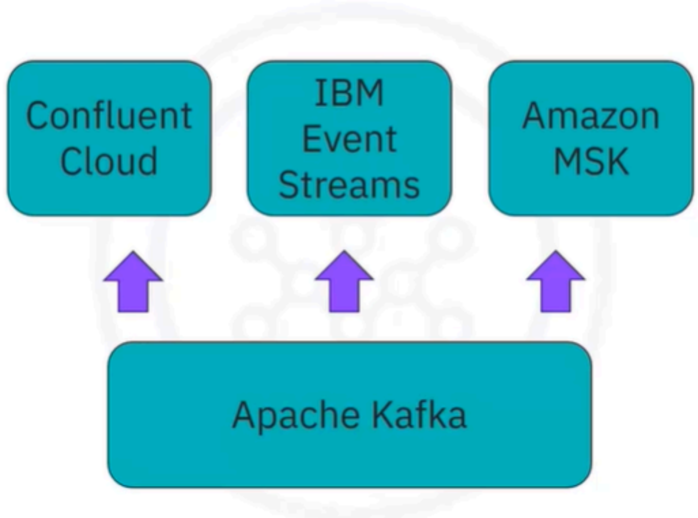
- A collection of shell scripts to communicate with the Kafka server, several high level programming APIs such as Java, Scala, Python, Go, C, and C++, rest APIs, and some specific third party clients made by the Kafka community. You can select different clients based on your requirements.



- Now that you have a basic understanding of Kafka, let's review the Kafka features. Kafka is a distribution system, which makes it highly scalable to handle high data throughput and concurrency. A Kafka cluster normally has multiple event brokers which can handle event streaming in parallel. Kafka is very fast and highly scalable.

- Kafka also divides event storage into multiple partitions and replications, which makes it fault-tolerant and highly reliable. Kafka stores the events permanently. As such, event consumption can be done whenever suitable for consumers without a deadline, and Kafka is open source, meaning that you can use it for free and even customize it based on your specific requirements.

Event streaming as a service



- Even though Kafka is open source and well documented, it is still challenging to configure and deploy Kafka without professional assistance. Deploying a Kafka cluster requires extensive efforts for tuning infrastructure and consistently adjusting the configurations, especially for enterprise-level deployments. Fortunately, several commercial service providers offer an on-demand ESP as a service to meet your streaming requirements.
- Many of them are built on top of Kafka and provide added value for customers. Some well known ESP providers include Confluent Cloud, which provides customers with fully managed Kafka services either on premises or on cloud. IBM event streams, which is also based on Kafka and provides many add-on services such as enterprise-grade security, disaster recovery, and 24/7 cluster monitoring. Amazon managed streaming for Apache Kafka, which is also a fully managed service to facilitate the build and deployment of Kafka.

Recap

In this video, you learned that:

- Apache Kafka is a popular open-source ESP
- Common Kafka use cases include user-activity tracking, metrics and log integrations, and financial transaction processing
- Apache Kafka is scalable and reliable
- Popular Kafka service providers include Confluent Cloud, IBM Event Stream, and Amazon MSK